MASTER THESIS · UPDATE 1

Grading handwritten exams with vision-language models

Like ETH’s STACK, which checks typed math by meaning, but for handwritten answers

[Your name] · Supervisor meeting · 28 October 2026

WHAT EXISTS

STACK checks meaning, but only for typed answers

TYPED + COMPUTER ALGEBRA

STACK, Numbas, WeBWorK, PrairieLearn

Grades by mathematical meaning (Maxima): *2(x+1) = 2x+2*. Randomised questions, instant feedback.

Limit: students must type math.

HANDWRITTEN + AI ASSIST

Gradescope (Turnitin)

AI groups similar handwritten answers; the instructor grades each group once.

Limit: a human still decides; fixed-template PDFs only.

RESEARCH 2025–26

Vision-language model graders

Levine et al. 2026: 95% rubric-item accuracy on university STEM work; most errors come from **misreading** handwriting.

Limit: no meaning check like STACK.

Sources: ethz.ch STACK page · Gradescope docs · arXiv 2605.19043, 2606.11477

DATASETS BY DIFFICULTY LEVEL

Public data exists for every level; math has the most

Level	Dataset	Contents	Use
Fill the gaps	Grabowski 2026	61 exams, 3,141 handwritten answer boxes	Evaluate
Short answers	CREG (German)	1,032 handwritten learner answers, 177 prompts	Evaluate
Free text	VisEssay (2022)	13k+ graded handwritten essays (no public download)	Evaluate
Math	MathWriting, HME100K	253k handwritten formulas → LaTeX; 99k photos	Fine-tune
Math	FERMAT (ACL 2025)	2,244 handwritten solutions with labelled errors	Evaluate
Math	CHECK-MAT (2025)	122 exam solutions with expert grades	Evaluate

Not found: a public German fill-the-gaps worksheet set. A 3,945-record calculus set (2026) is not released.

PROPOSED APPROACH

Fine-tune the reading step, then check like STACK

1 · Capture

Scan or photo of the answer sheet

2 · Read

Vision-language model → text / LaTeX. Fine-tune here.

3 · Check

SymPy equivalence, as in STACK; LLM for rubric text

4 · Grade

Score and feedback per step, teacher reviews

Why step 2

For one model, 87% of grading errors came from misreading handwriting (Levine et al. 2026).

Open models to compare

Qwen2.5-VL, olmOCR 2, PaddleOCR-VL, DeepSeek-OCR, Chandra OCR 2

Reference points

Gemini, Mistral OCR (closed APIs); VEHME (fine-tuned grader, EMNLP 2025)

DRAFT RESEARCH QUESTION

Does fine-tuning an open vision-language model on handwritten math improve grading accuracy when its output is checked by computer algebra, STACK-style?

Hypothesis: fine-tuned reading + algebra check beats a zero-shot model grading directly, measured on FERMAT.

NEXT TWO WEEKS

1. Download FERMAT and MathWriting; check licences
2. Zero-shot baseline: Qwen2.5-VL vs Gemini on a FERMAT subset
3. Prototype the SymPy equivalence check
4. Read 5 core papers closely, skim about 20

QUESTIONS FOR YOU

- **Tasks:** math + fill-the-gaps with synthetic handwriting – OK?
- **Essays:** drop, or ask the VisEssay authors for data?
- **Seminar paper:** VEHME or Henkel et al.?